# U.S. Trade Flow Composition Analysis Visualization

(2020-2024)

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# Project Objectives

1. Analyze the composition of U.S. trade flows and their economic impact.
2. Identify major trading partners and emerging market opportunities.
3. Assess the centrality of U.S. ports in the global trade network.
4. Highlight potential vulnerabilities and strategic growth areas.

# Data Overview

Years covered: 2020 to 2024

Includes U.S. District and Foreign Country data

Trade metrics: Value (USD)

Commodities group by 21 Sections

Data sources: DATAWEB https://dataweb.usitc.gov/trade/search/Export/HTS

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Data Type | Country | Year | HTS Number | Description | District | Value | Flow | Section |
| 2 | FAS Value | Afghanistan | 2020 | 94 | FURNITURE; BEDDING, CUSHIONS ETC. | Washington, DC | 35341 | DomesticExports | 20 |
| 3 | FAS Value | Afghanistan | 2020 | 82 | TOOLS, IMPLEMENTS, CUTLERY, SPOON | Cleveland, OH | 631593 | DomesticExports | 15 |
| 4 | FAS Value | Afghanistan | 2020 | 84 | NUCLEAR REACTORS, BOILERS, MACHIN | Charlotte, NC | 4250 | DomesticExports | 16 |
| 5 | FAS Value | Afghanistan | 2020 | 82 | TOOLS, IMPLEMENTS, CUTLERY, SPOON | Chicago, IL | 465151 | DomesticExports | 15 |
| 6 | FAS Value | Afghanistan | 2020 | 82 | TOOLS, IMPLEMENTS, CUTLERY, SPOON | Charleston, SC | 3637 | DomesticExports | 15 |
| 7 | FAS Value | Afghanistan | 2020 | 61 | ARTICLES OF APPAREL AND CLOTHING | Baltimore, MD | 6848 | DomesticExports | 11 |
| 8 | FAS Value | Afghanistan | 2020 | 61 | ARTICLES OF APPAREL AND CLOTHING | Chicago, IL | 69721 | DomesticExports | 11 |
| 9 | FAS Value | Afghanistan | 2020 | 61 | ARTICLES OF APPAREL AND CLOTHING | Cleveland, OH | 50592 | DomesticExports | 11 |
| 10 | FAS Value | Afghanistan | 2020 | 84 | NUCLEAR REACTORS, BOILERS, MACHIN | San Francisco, CA | 43774 | DomesticExports | 16 |

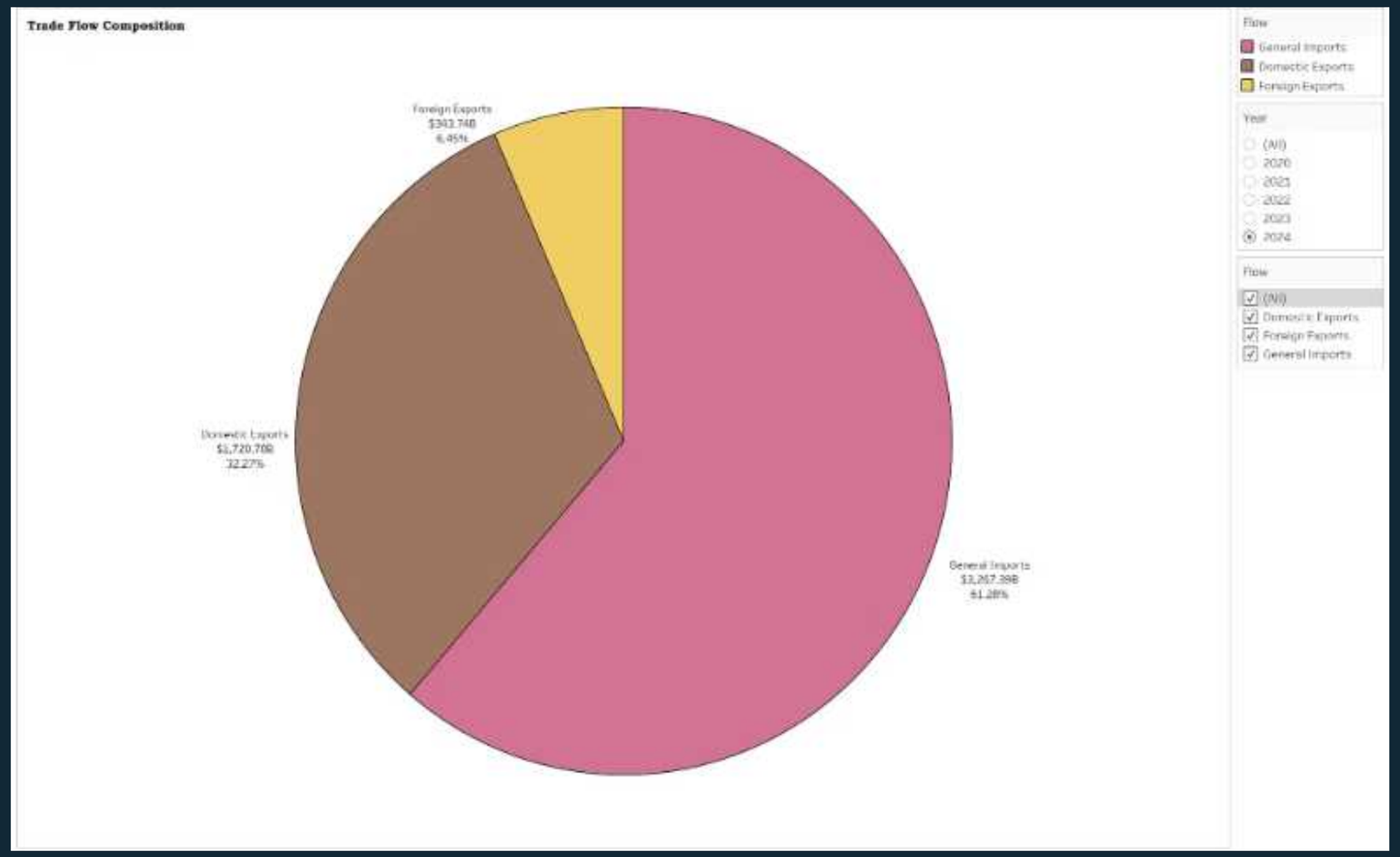
| Section | Name | HS Chapters |
|---|---|---|
| 1 | Live Animals; Animal Products | 1-5 |
| 2 | Vegetable Products | 6-14 |
| 3 | Animal or Vegetable Fats and Oils | 15 |
| 4 | Prepared Foodstuffs; Beverages, Spirits, Tobacco | 16-24 |
| 5 | Mineral Products | 25-27 |
| 6 | Products of the Chemical or Allied Industries | 28-38 |
| 7 | Plastics, Rubber and Articles Thereof | 39-40 |
| 8 | Raw Hides, Skins, Leather | 41-43 |
| 9 | Wood and Articles of Wood | 44-46 |
| 10 | Pulp of Wood, Paper | 47-49 |
| 11 | Textile and Textile Articles | 50-63 |
| 12 | Footwear, Headgear, etc. | 64-67 |
| 13 | Articles of Stone, Plaster, Cement, Ceramics, Glass | 68-70 |
| 14 | Pearls, precious stones, precious metals | 71 |
| 15 | Base Metals and Articles of Base Metal | 72-83 |
| 16 | Machinery, Appliances, Electrical Equipment | 84-85 |
| 17 | Vehicles, Aircraft, Vessels | 86-89 |
| 18 | Precision Instruments | 90-92 |
| 19 | Arms and Ammunitions | 93 |
| 20 | Misc. Manufactured Articles | 94-96 |
| 21 | Works of Art | 97 |
| 22 | Special Classification Provisions | 98-99 |

Notes: This table displays sections and chapters U.S. Import HS Codes. Section names have been shortened for brevity. See the website of the U.S. International Trade Commission for full section names.

# U.S. Trade Flow Composition(2020-2024)

- Three main categories:
  - General Imports (Pink) : **61.28%**
  - Domestic Exports (Brown) : 32.27%
  - Foreign Exports (Yellow) : 6.45%

- Insights: Trade deficits, currency pressure

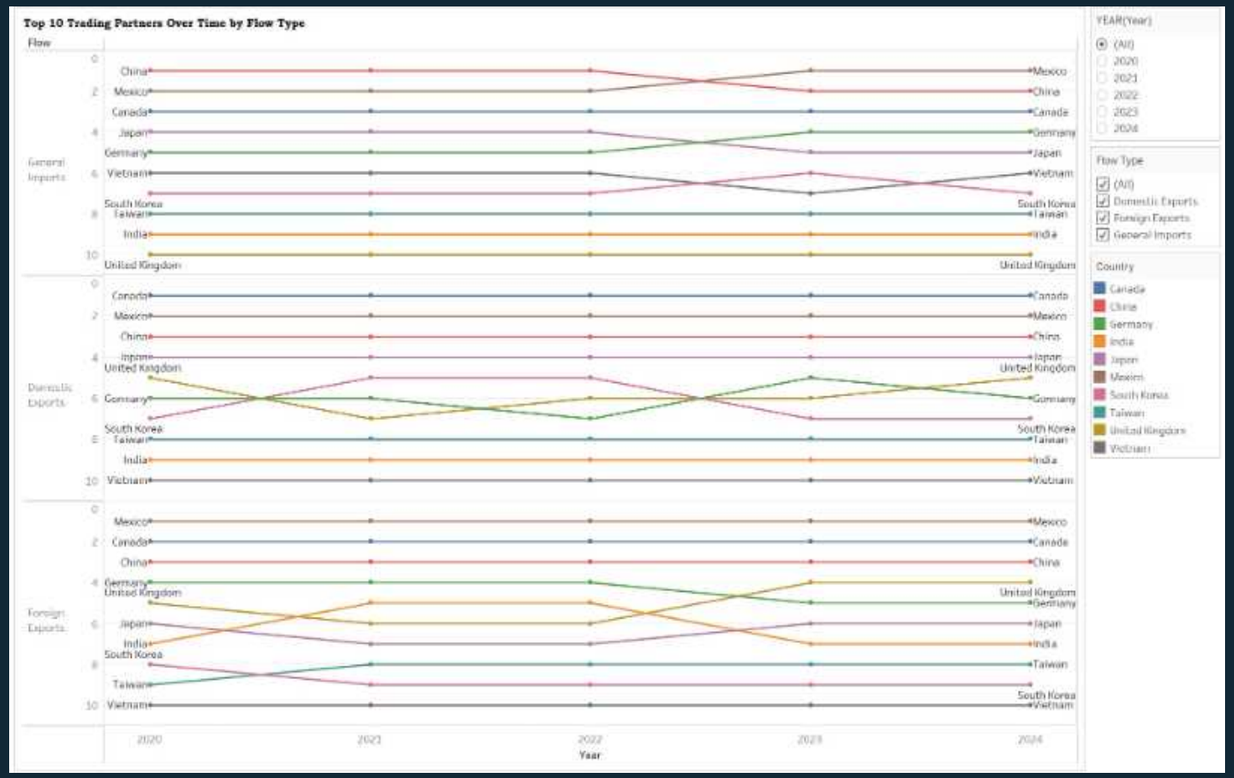


# U.S. Trade Value Distribution by Country & Flow Type

- The size of each rectangle represents the relative share of the total trade value contributed by each country within each flow type.
- The percentages indicate the proportion of the overall trade value for each flow category.
- Concentration of Trade: The largest blocks are dominated by a few countries, reflecting the U.S.'s dependence on specific major trading partners.

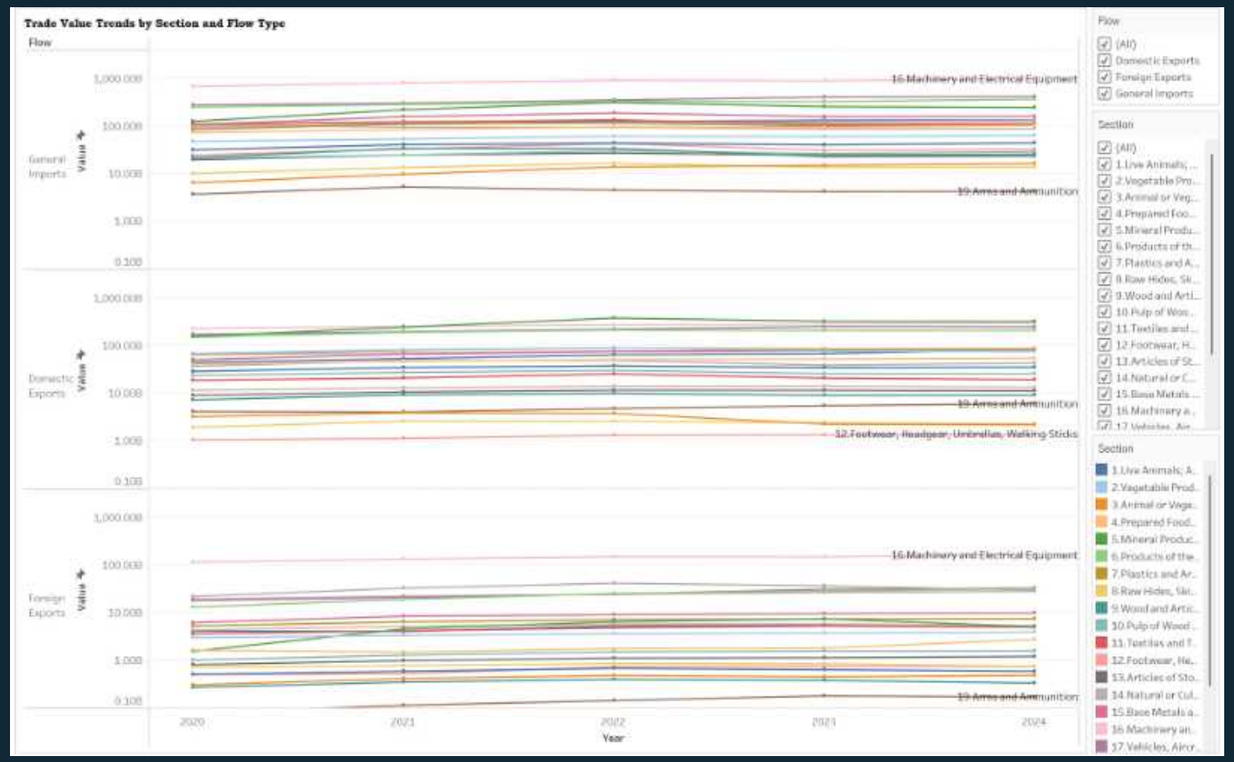
# Top 10 Trading Partners Over Time by Flow Type

- The Top 10 U.S. Trading Partners across three different flow types from 2020 to 2024.
- The lines connecting the dots show how the rank of each trading partner has shifted over the five-year period.
- The ranking is based on the total trade value within each category, with the most significant trading partners appearing closer to the top.
- Supply Chain Risks: Heavy reliance on a few key partners, especially in imports, presents risks during geopolitical tensions or supply chain disruptions.



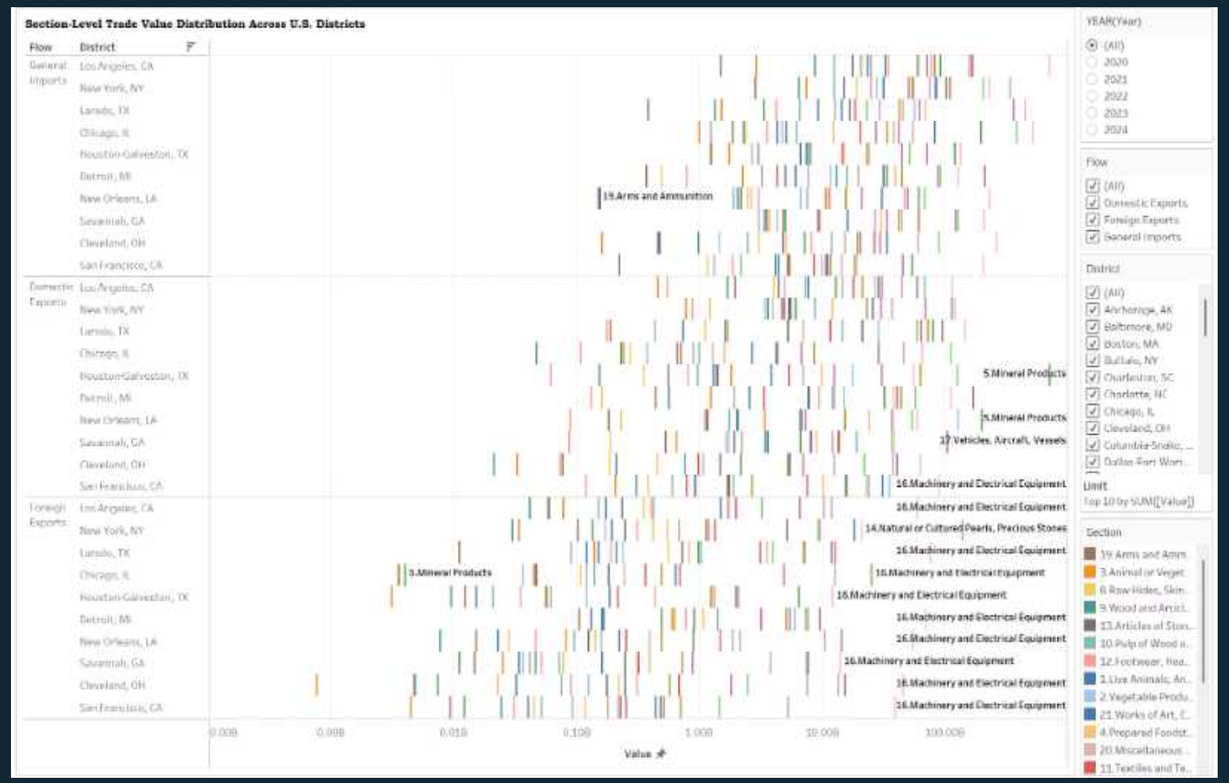
# U.S. Trade Value Trends by Section and Flow Type

- Each line represents a specific commodity section.
- The chart uses a logarithmic scale to capture the wide range of trade values, from small categories like Arms and Ammunition to massive sectors like Machinery and Electrical Equipment.
- Consistent Leaders: Machinery and Electrical Equipment (Section 16) consistently leads across all flow types, indicating its critical role in both imports and exports.



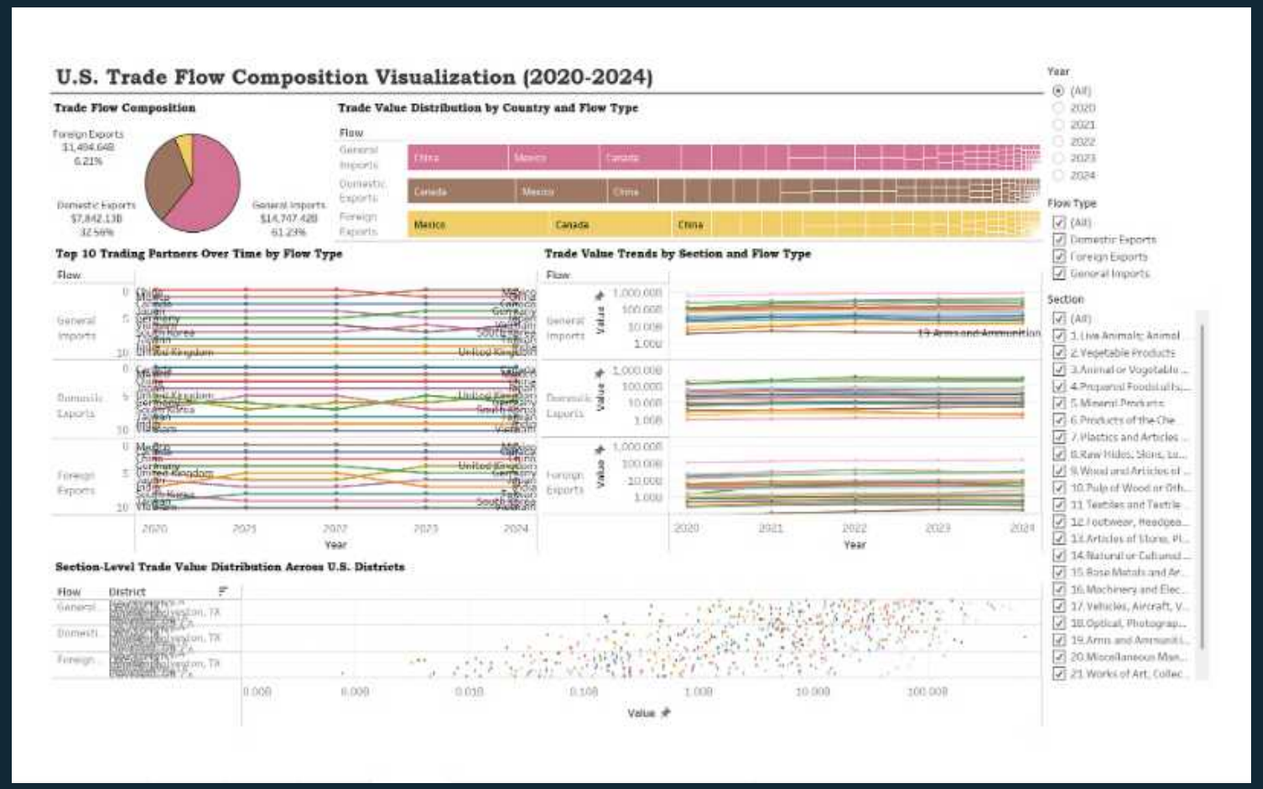
# Section-Level Trade Value Distribution Across U.S. Districts

- This Gantt Bar Chart shows the distribution of trade values for different Sections across U.S. customs Districts and Flow Types.
- The horizontal axis represents the Trade Value on a logarithmic scale, while the vertical axis is divided by Flow Type and District.
- Each colored bar represents a distinct section, with the color legend on the right indicating the specific commodity category.
- Major districts like Los Angeles, CA, New York, NY, and Houston-Galveston, TX display a wide distribution of trade values across many sections.



# U.S. Trade Flow Dashboard (2020–2024)

- The dashboard effectively captures the scale and diversity of U.S. trade.
- The combination of Line charts, Tree map, and Gantt bar chart provide a multi-dimensional view of trade trends, supporting both high-level overviews and detailed exploration.
- Allows dynamic filtering by Year, Flow Type, and Section to refine the analysis.
- Provides flexibility to isolate specific timeframes, commodity categories, and trade directions.



# Graph Network Analysis Visualization

## General Imports

- Nodes:

  Country: Canada, Mexico

  District: New York, NY; Los Angeles, CA

- Edges:

  source: Country (Canada, Mexico)

  target: District (New York, NY; Los Angeles, CA)

- Weights:

  Value in Dollars ($)

## Domestic Exports / Foreign Exports

- Nodes:

  Country: Canada, Mexico

  District: New York, NY; Los Angeles, CA

- Edges:

  source: District (New York, NY; Los Angeles, CA)

  target: Country (Canada, Mexico)

- Weights:

  Value in Dollars ($)

# U.S. Trade Network - Centrality by Flow and Section

4

## Year

2020 - 2024

4

## Flow Type

General Imports, Domestic Exports, Foreign Exports with all

22

## Section

21 Sections with All

4

## Centrality

degree, betweenness, closeness, pagerank

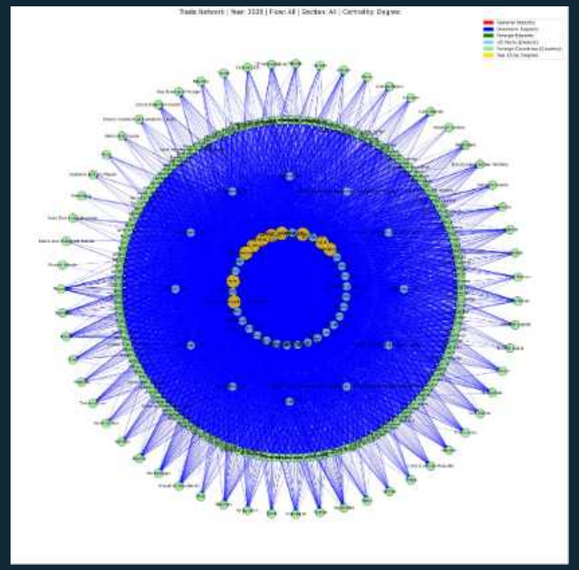
1408

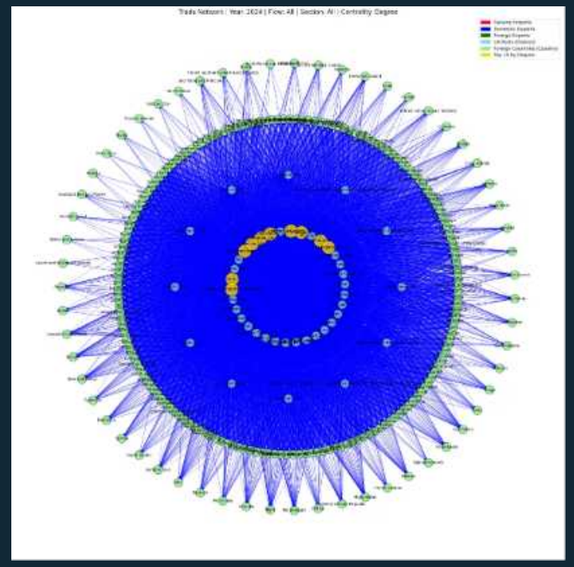
## Combination results

Graph Network

# U.S. Trade Network

## - All Flows and Sections & Degree Centrality (2020 and 2024)





Year: 2020 | Flow: All | Section: All | Metric: Degree

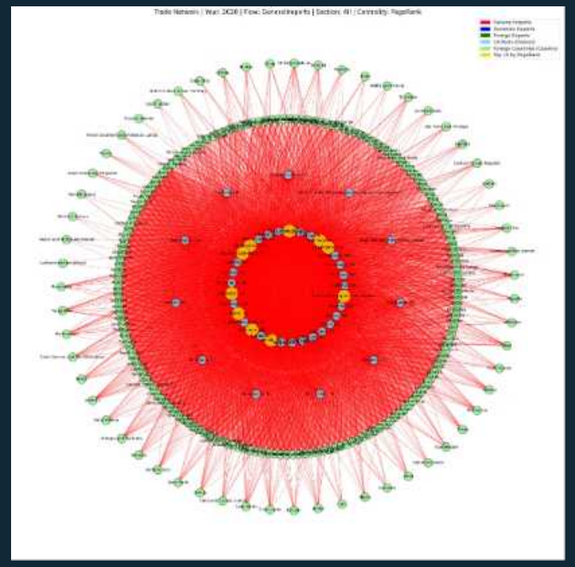
| | Node | Degree |
|---|---|---|
| 31 | Low-valued imports and exports | 1.629496402877698 |
| 10 | New York, NY | 1.60431654676259 |
| 2 | Cleveland, OH | 1.589928057553957 |
| 8 | Los Angeles, CA | 1.571942446043165 5 |
| 9 | New Orleans, LA | 1.564748201438849 |
| 23 | Miami, FL | 1.496402877697841 7 |
| 4 | Chicago, IL | 1.464028776978417 3 |
| 14 | Houston-Galveston, TX | 1.464028776978417 3 |
| 7 | San Francisco, CA | 1.438848920863309 5 |
| 11 | Savannah, GA | 1.413669064748201 5 |

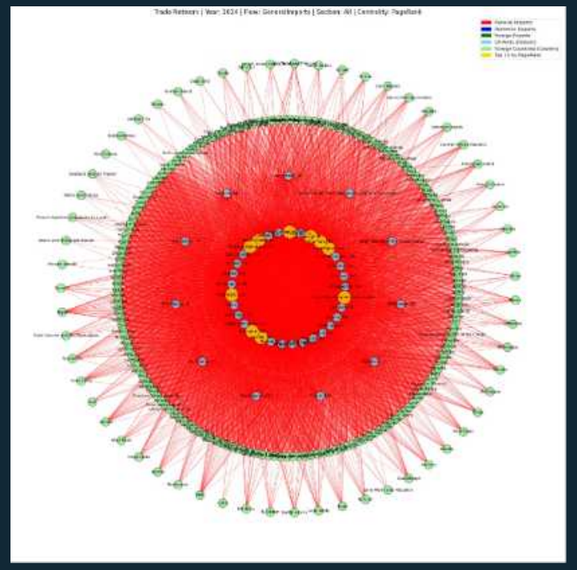
Year: 2024 | Flow: All | Section: All | Metric: Degree

| | Node | Degree |
|---|---|---|
| 17 | Low-valued imports and exports | 1.625899280575539 7 |
| 3 | New York, NY | 1.597122302158273 4 |
| 5 | Cleveland, OH | 1.575539568345323 7 |
| 11 | Los Angeles, CA | 1.564748201438849 |
| 10 | Miami, FL | 1.557553956834532 5 |
| 9 | New Orleans, LA | 1.550359712230216 |
| 12 | Chicago, IL | 1.482014388489208 7 |
| 0 | Savannah, GA | 1.482014388489208 7 |
| 15 | Houston-Galveston, TX | 1.460431654676259 |
| 2 | San Francisco, CA | 1.438848920863309 5 |

# U.S. Trade Network

## - General Imports & All Sections & PageRank Centrality (2020 and 2024)





Year: 2020 | Flow: GeneralImports | Section: All | Metric: PageRank

| | Node | PageRank |
|---|---|---|
| 20 | Low-valued imports and exports | 0.10059428602125277 |
| 1 | New York, NY | 0.03159400597157695 |
| 3 | Los Angeles, CA | 0.02259845913599415 5 |
| 6 | Miami, FL | 0.021381402366072 18 |
| 27 | Savannah, GA | 0.01875718226426709 |
| 15 | Houston-Galveston, TX | 0.01861962587649535 6 |
| 28 | Baltimore, MD | 0.01853038997610121 5 |
| 10 | San Francisco, CA | 0.01733812986274736 7 |
| 25 | Charleston, SC | 0.01651989801580234 7 |
| 22 | Norfolk, VA | 0.01556191696684920 2 |

Year: 2024 | Flow: GeneralImports | Section: All | Metric: PageRank

| | Node | PageRank |
|---|---|---|
| 21 | Low-valued imports and exports | 0.10298939062611619 |
| 16 | New York, NY | 0.03140461109019214 |
| 5 | Baltimore, MD | 0.02298108173049198 |
| 26 | Charleston, SC | 0.01854550654312211 7 |
| 4 | Norfolk, VA | 0.01767886249308178 4 |
| 7 | New Orleans, LA | 0.01698471512757474 7 |
| 1 | Los Angeles, CA | 0.01593786237071 84 |
| 8 | Savannah, GA | 0.01506805735043237 9 |
| 20 | Houston-Galveston, TX | 0.01505474156234035 7 |
| 25 | Chicago, IL | 0.01443583714379457 |

# U.S. District General Imports Network

1. Build a District-Section Value Matrix

| Section | 1 | 2 | 3 | 4 | 5 | \ |
|---|---|---|---|---|---|---|
| District | | | | | | |
| Anchorage, AK | 7943814 | 2975658 | 42983 | 5127700 | 731932268 | |
| Baltimore, MD | 441060517 | 971660781 | 157697440 | 1701044483 | 363985832 | |
| Boston, MA | 695476272 | 65379052 | 136653882 | 561927422 | 3033078476 | |
| Buffalo, NY | 548014322 | 783494710 | 88510056 | 2678604043 | 2240084419 | |
| Charleston, SC | 91835047 | 666145598 | 91817514 | 536680272 | 283798857 | |

| Section | 6 | 7 | 8 | 9 | 10 | ... | \ |
|---|---|---|---|---|---|---|---|
| District | | | | | | ... | |
| Anchorage, AK | 347720390 | 101651259 | 10170608 | 9136219 | 52498376 | ... | |
| Baltimore, MD | 2599796536 | 897432058 | 54532507 | 882999278 | 740968075 | ... | |
| Boston, MA | 1394956363 | 680567385 | 59640334 | 116737555 | 150536656 | ... | |
| Buffalo, NY | 3237912916 | 2215635308 | 55167959 | 509660328 | 989941791 | ... | |
| Charleston, SC | 6151256994 | 3704275473 | 96951549 | 536107798 | 337323360 | ... | |

2. Calculate Cosine_Similarity

| District | Anchorage, AK | Baltimore, MD | Boston, MA | Buffalo, NY | \ |
|---|---|---|---|---|---|
| District | | | | | |
| Anchorage, AK | 1.000000 | 0.348771 | 0.656321 | 0.564642 | |
| Baltimore, MD | 0.348771 | 1.000000 | 0.489245 | 0.875242 | |
| Boston, MA | 0.656321 | 0.489245 | 1.000000 | 0.737062 | |
| Buffalo, NY | 0.564642 | 0.875242 | 0.737062 | 1.000000 | |
| Charleston, SC | 0.817665 | 0.681268 | 0.727441 | 0.853269 | |

| District | Charleston, SC | Charlotte, NC | Chicago, IL | Cleveland, OH | \ |
|---|---|---|---|---|---|
| District | | | | | |
| Anchorage, AK | 0.817665 | 0.583258 | 0.858809 | 0.935476 | |
| Baltimore, MD | 0.681268 | 0.341529 | 0.378755 | 0.417635 | |
| Boston, MA | 0.727441 | 0.601635 | 0.743526 | 0.742526 | |
| Buffalo, NY | 0.853269 | 0.623932 | 0.676876 | 0.667679 | |
| Charleston, SC | 1.000000 | 0.778551 | 0.879488 | 0.921868 | |

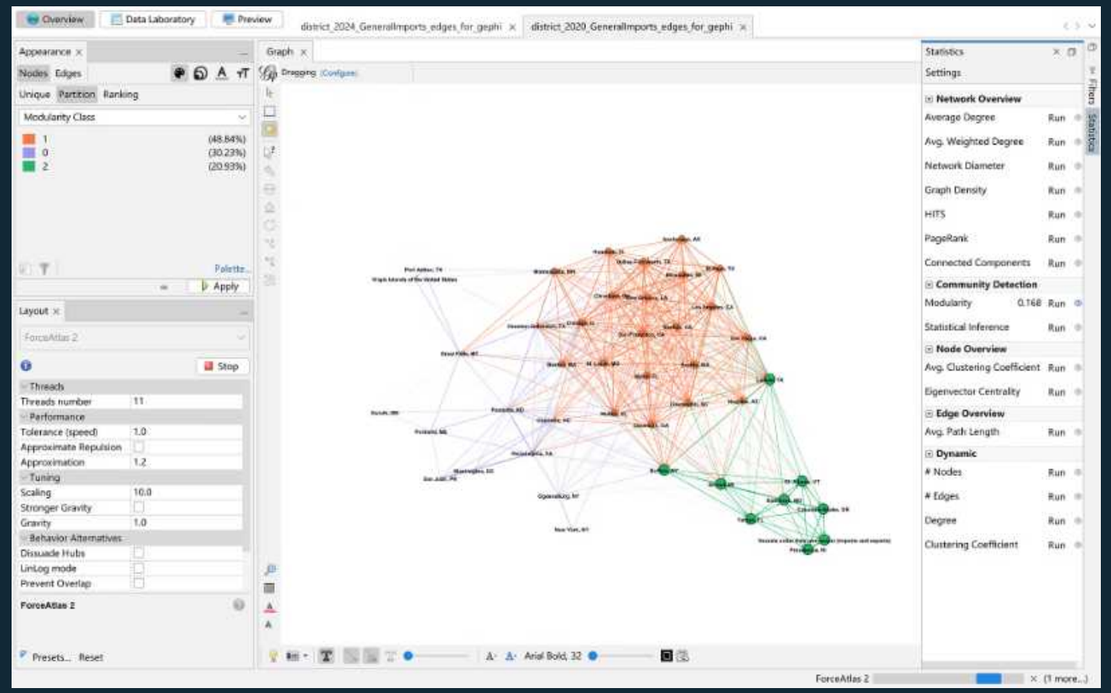
| District | Columbia-Snake, OR | Dallas-Fort Worth, TX | ... | \ |
|---|---|---|---|---|
| District | | | ... | |
| Anchorage, AK | 0.254345 | 0.981614 | ... | |
| Baltimore, MD | 0.973079 | 0.387761 | ... | |
| Boston, MA | 0.421392 | 0.703409 | ... | |
| Buffalo, NY | 0.776526 | 0.635446 | ... | |
| Charleston, SC | 0.569204 | 0.881952 | ... | |

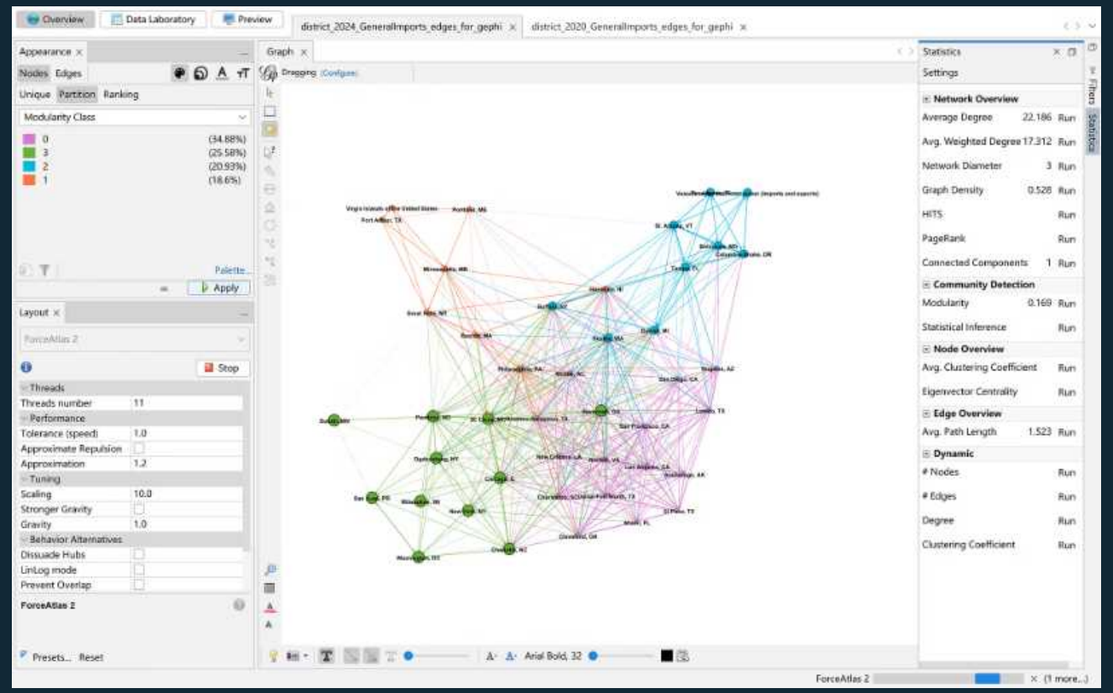
3. Convert the Matrix to an Edges List

| | Source | Target | Weight |
|---|---|---|---|
| 0 | Anchorage, AK | Boston, MA | 0.656321 |
| 1 | Anchorage, AK | Charleston, SC | 0.817665 |
| 2 | Anchorage, AK | Chicago, IL | 0.858809 |
| 3 | Anchorage, AK | Cleveland, OH | 0.935476 |
| 4 | Anchorage, AK | Dallas-Fort Worth, TX | 0.981614 |

# U.S. District General Imports Network

## - Clustered by Section Similarity (2020 vs 2024)



# Conclusion

This analysis highlights key trends in U.S. trade flows and network dynamics.

Understanding trade value distribution and centrality helps optimize strategic decisions.

Visual insights into trade relationships guide policy and economic impact assessments.

## Key Insights

Trade network centrality reveals pivotal ports and countries in U.S. commerce.

## Data-Driven Decisions

Trade flow composition informs tariff impact and economic strategy adjustments.

## Future Outlook

Continued monitoring ensures adaptation to global trade shifts and policy changes.